Juan Andrés Herrera Vega
Backend Software Developer | Systems, Automation, Reliability
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Summary

Backend developer focused on systems, automation, monitoring utilities, and data pipelines across Linux and Windows environments. Experienced building reliable software from problem definition through deployment, with work spanning research pipelines, operational data processing, and infrastructure-oriented tooling.

Experience

Software Developer – Client Work | Nov 2025 – Present

• Built and maintain a Python data-processing pipeline that ingests evolving incident datasets, normalizes inputs, and produces standardized reporting outputs.

• Implemented validation, data normalization, and automated report generation.

Software Developer – Research Team at Brain Lab, Concordia University | Sept 2024 – Dec 2024

• Developed automated MATLAB processing pipelines and modules for a Quantitative Susceptibility Mapping toolbox used in MRI imaging workflows.

• Integrated 10+ imaging and metadata sources with validation and error logging, improving pipeline reliability and reproducibility of outputs.

Computer Science Tutor | Sep 2021 – Dec 2023

• Mentored 5 students in Java, Python, C++, C#, Object-Oriented Programming, and application development

Projects

Chi-Separation Software – Medical Imaging Toolbox | Team Project (Sep 2024 – Jan 2025)

• Developed MATLAB toolbox for quantitative susceptibility mapping from MRI data, improving analysis accuracy across 30+ scans and supporting early Alzheimer's detection research.

• ***Tools:*** *MATLAB, Image Processing Toolbox, Google Colab, Git, dcm2niix, Python*

ShielD | Personal Project (May 2024 – Nov 2024)

• Built lightweight cybersecurity utility to monitor web traffic integrity and detect unauthorized access patterns.

• ***Tools:*** *Java, TensorFlow Java, Tablesaw, Maven, Git*

InfrawatchLite | Personal Project (Dec 2024 - Feb 2025)

• Built Linux monitoring tool that triggers alerts when CPU/GPU usage exceeds thresholds, supporting system observability and early intervention.

• ***Tools:*** *Python, PyTorch, Git, Bash, NumPy, Pandas, Matplotlib*

HelixNet | Personal Project (Feb 2025 – July 2025)

• Built a Machine Learning model trained to classify protein structures with a full pipeline marking 89% accuracy.

• ***Tools:*** *Java, TensorFlow Java, Weka, Maven, Git*

Skills and Certifications

Certifications: Linux Foundation Certified Systems Administrator (LFCS) **|** In Progress, Oct 2025.

Programming: Python, Java, C/C++, Bash, PowerShell, Node.js, C#, Visual IDE

Tools and Frameworks: SQL/MySQL, REST APIs, Pandas, NumPy, scikit-learn, TensorFlow, Git

Systems and Security: Linux and Windows administration, Active Directory, IAM, file integrity monitoring

Languages: English (Native), Spanish (Native), French (Intermediate)

Education

Bachelor of Science with double major – Concordia University, Montreal, Quebec

Mathematics and Statistics **(Focus in Computer Science)**

Physics **(Specialization in Biophysics)**

Awards: MES COVID Bursary (2020/2021), Campaign New Millennium Scholarship (2021/2022)